

Evaluation of Turner relaxed state as a model of long-lived ion-trapping structures in plasma focus and Z-pinches

S. K. H. Auluck

Advanced Technology Systems Section, Bhabha Atomic Research Center, Mumbai 400 085, India

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Relatively long-lived spheroidal structures coincident with the neutron emission phase have been observed in frozen deuterium fiber Z-pinch and some plasma focus devices. Existence of energetic ion-trapping mechanism in plasma focus has also been inferred from experimental data. It has been conjectured that these are related phenomena. This paper applies Turner's theory [L. Turner, IEEE Trans. Plasma Sci. **14**, 849 (1986)] of relaxation of a Hall magnetofluid to construct a model of these structures and ion-trapping mechanism. Turner's solution modified for a finite-length plasma is used to obtain expressions for the magnetic field, velocity, and equilibrium pressure fields and is shown to represent an entity which is simultaneously a fluid vortex, a force-free magnetic field, a confined finite-pressure plasma, a charged object, and a trapped energetic ion beam. Characteristic features expected from diagnostic experiments are evaluated and shown to resemble experimental observations. © 2011 American Institute of Physics. [doi:10.1063/1.3554701]

I. INTRODUCTION

The emission of neutrons from Z-pinches^{1,2} and plasma focus devices³⁻⁵ has some characteristic features which can be understood only in terms of a model where energetic ions (~ 100 keV) are trapped inside a small volume for a long time. The essential experimental facts which point toward such model are:

- The reacting ions responsible for neutron emission have a kinetic energy in the range from 40 (Ref. 1) to 600 (Ref. 2) keV (as inferred from neutron spectra) for which the measured or estimated plasma density and temperature imply a fusion reaction mean free path several orders of magnitude larger than the size of the reaction zone measured using reaction product or luminosity imaging.
- The duration of neutron emission is much larger than the energetic ion transit time across the reaction zone.
- Particle and energy content estimates derived from experimental data are grossly at variance with the assumption that ions are continuously generated and produce fusion reaction during a single transit across the plasma.^{4,5}

Correlation has been observed between neutron emission and appearance of luminous plasma structures,^{1,3,5} distinguishable from the gross plasma by a well-discernible boundary. In a frozen deuterium Z-pinch,¹ these take the form of "beads" on a string. Isolated single spheroidal structures are observed on plasma focus devices.^{3,5} Conjectures have been made identifying these structures with the ion-trapping mechanism.^{3,5}

Sometimes, the neutron emission observed 90° to the axis is more than that observed in the direction of current (at 0°).⁴ Space-resolved, time integrated spectra of neutrons⁵ and reaction protons show⁶ a shift of opposite sign at mirror-symmetric lateral coordinates suggesting a rotating population of reacting deuterons. Presence of axial magnetic field in the pinch phase has been inferred using ion-optics simulation

from radial shift of the reaction proton image with respect to neutron image in the POSEIDON plasma focus⁶ and from Faraday rotation in Frascati plasma focus.⁵

Achieving symmetrical plasma formation is usually the main technical objective of plasma focus construction and azimuthally symmetric plasma designed to be driven with an axial current density is not expected to possess either axial magnetic field or rotational features. Clearly, a topological change in the plasma-magnetic field system occurs along with significant transfer of energy to plasma ions. An associated feature of this topological change is the growth of $m=0$ mode.

A significant clue to the process of energy transfer to the ions is found in the work on nonlinear development of the $m=0$ mode by Vikhrev and coworkers (reviewed in Ref. 2), which reveals the low density plasma surrounding the dense core of the Z-pinch to be a source of fast ions. The $m=0$ instability creates a neck region and the associated axial convective flow redistributes plasma density^{7,8} to create a "cavity," which is rapidly filled with the low density outlying plasma at a speed of the order of Alfvén velocity, V_A . The presence of a large population of energetic ions can thus be accounted for by an *a priori* assumption that the plasma-magnetic field system is displaced by $m=0$ mode. The presence of axial magnetic field and rotational features of the population of energetic ions is however still beyond the reach of this model.

It is well known^{9,10} that plasma relaxation in the presence of Hall effect leads to states with significant mass flow (particularly rotation) on the scale of Alfvén velocity and possessing axial magnetic field even for azimuthally symmetric case. In this context, it has been recently suggested¹¹ that the low density current carrying sheath in a plasma focus could be described as Turner relaxed state (TRS) of an infinitely long annular plasma cylinder. If this "infinitely long" plasma were assumed to be broken into regions of finite axial extent following $m=0$ instability, the resulting model could

reproduce many of the experimental observations mentioned above.

The problem of plasma relaxation in the presence of Hall effect has many facets being actively investigated both analytically and numerically. The question of nature of constraints on the evolution of the plasma has been considered in different ways by many investigators. Sudan¹⁰ has used three-components of canonical momenta, magnetic helicity and total mass as invariants. Turner⁹ uses magnetic helicity and hybrid helicity, which is shown to be an independent integral invariant of the Hall magnetohydrodynamics (MHD) dissipationless dynamics. Steinhauer and Ishida argue that total energy should decay faster than the two helicities and should be an appropriate target functional for constrained minimization.¹² Yoshida and Mahajan, on the other hand, propose minimizing squared modulus of the generalized enstrophy (curl of the canonical momentum) while keeping energy, magnetic helicity, and helicity of generalized vorticity constant.¹³ Khalzov, Ebrahimi, Schnack, and Mirnov use axial magnetic flux, axial momentum and angular momentum as invariants in addition to magnetic helicity and cross helicity and minimize energy.¹⁴ Di Vita obtains a generalized variational principle from which Turner's relaxation theory and many other phenomena can be obtained as special cases.¹⁵

The form of the self-organized plasma state is also a topic of interest. Taylor's ideal MHD relaxed state is a force-free magnetic field, with zero pressure, described by a single curl eigenfunction.¹⁶ Turner's relaxed state is composed of two curl eigenfunctions.⁹ Yoshida *et al.* investigate the properties of the double-Beltrami structure of the tightly coupled kinematic and electromagnetic properties of the Hall magnetofluid.¹⁷ Ohsaki and Yoshida have considered stability properties of these double-Beltrami states.¹⁸ Dasgupta *et al.* derive a *triple*-Beltrami form for the relaxed state by minimizing the Ohmic energy dissipation rate subject to constancy of magnetic helicity.¹⁹ The role of Hall MHD in formation of filaments, also conjectured to be relaxed states¹⁵ has been elucidated in the numerical simulations of Dreher *et al.*²⁰

This paper attempts to examine the possibility of understanding the experimental results mentioned above by applying Turner's solution⁹ to a finite cylinder. Turner's theory is chosen because it is a special case of many other theories and appears to be the simplest relaxation model consistent with Hall MHD. The focus of this paper is on understanding the properties of the finite TRS as a mathematical object. As with all relaxation theories, Turner's theory attempts to pro-

vide a reasonable approximation to real plasma in a limited context, namely, the context of incompressible flow. In an experimental context, however, one may legitimately inquire about the conditions of validity of the approximation and the physical implications of the neglected phenomena. Such inquiry forms part of the discussion of this paper.

II. GENERAL PROPERTIES OF TRS

This section looks at properties of Turner's solution (whose degenerate limit for the case of an infinitely long annular cylinder was discussed in Ref. 11) when applied to the case of finite plasma. Recapitulating the definitions: m =average ion mass, ρ =mass density, n =ion number density, c =speed of light, $\vec{B}, \vec{v}, \vec{R}$ denote magnetic field, velocity and position vector, $\vec{\omega}_c \equiv e\vec{B}/m$; $\omega_p^2 \equiv (ne^2/\epsilon_0 m)$, $\vec{\omega} \equiv \vec{\nabla} \times \vec{v}$; $\vec{R} \equiv (\omega_p/c)\vec{R}$; $(\omega_p/c)\vec{v} \equiv \vec{V}$; $\vec{V} \times \vec{Z} = \tilde{\mu}_+ \vec{Z}_+ - \tilde{\mu}_- \vec{Z}_-$, $1 + \alpha_- = -(1/2)\tilde{\mu}_+ \{\tilde{\mu}_+ - \tilde{\mu}_- - \sqrt{(\tilde{\mu}_+ - \tilde{\mu}_-)^2 + 4}\}$, $1 + \alpha_+ = (1/2)\tilde{\mu}_- \{\tilde{\mu}_+ - \tilde{\mu}_- + \sqrt{(\tilde{\mu}_+ - \tilde{\mu}_-)^2 + 4}\}$.

$\vec{Z}_{\pm} \equiv C_{\pm} \vec{Z}(\vec{R}, \tilde{\mu}_{\pm})$ the principal results from Turner's theory are

$$\vec{B} = B_0 \frac{\vec{Z}_+ - \vec{Z}_-}{\alpha_+ - \alpha_-} \quad (1)$$

$$\vec{J} = B_0 \frac{\omega_p}{\mu_0 c} \frac{\tilde{\mu}_+ \vec{Z}_+ - \tilde{\mu}_- \vec{Z}_-}{\alpha_+ - \alpha_-} \quad (2)$$

$$\vec{v} = \frac{1}{2} \frac{c}{\omega_p} \frac{e B_0}{m} \frac{1}{(\alpha_+ - \alpha_-)} \times \{(\tilde{\mu}_+ - \tilde{\mu}_-)(\vec{Z}_+ + \vec{Z}_-) - (\sqrt{(\tilde{\mu}_+ - \tilde{\mu}_-)^2 + 4})(\vec{Z}_+ - \vec{Z}_-)\}. \quad (3)$$

Since Turner's theory is applicable to a steady state (constant density), the irrotational component of velocity is zero. The momentum equation then yields the Bernoulli condition,¹⁰ after substitution from Eqs. (1)–(3),

$$\vec{\nabla} \left(p + \frac{1}{2} \rho |\vec{v}|^2 \right) = \vec{J} \times \vec{B} + \rho \vec{v} \times \vec{\nabla} \times \vec{v} = 0. \quad (4)$$

The solution of Eq. (4) is

$$p(\vec{R}) = p_0 - \frac{1}{2} \rho \vec{v} \cdot \vec{v} = p_0 - \frac{B_0^2}{2\mu_0} \tilde{p}(\vec{R}) \geq 0 \quad (5)$$

$$\tilde{p}(\vec{R}) \equiv \left\{ \frac{(\tilde{\mu}_+ - \tilde{\mu}_-)(\vec{Z}_+ + \vec{Z}_-) - (\sqrt{(\tilde{\mu}_+ - \tilde{\mu}_-)^2 + 4})(\vec{Z}_+ - \vec{Z}_-)}{2(\alpha_+ - \alpha_-)} \right\}^2. \quad (6)$$

Turner's theory neglects electron pressure in the generalized Ohm's Law. The consequences of this approximation can be examined by assuming the electron pressure to be related to the total pressure p as $p_e = Z_{\text{eff}} p / (Z_{\text{eff}} + 1)$, Z_{eff} being the effective atomic number. The generalized Ohm's Law then yields a nonsolenoidal electric field and electrostatic potential φ for the steady state

$$\begin{aligned}\vec{E} &= -\vec{v} \times \vec{B} + \frac{1}{en} \{ \vec{J} \times \vec{B} - \vec{\nabla} p_e \} \\ &= -\frac{1}{en} \vec{\nabla} p_e = \frac{B_0}{\sqrt{\mu_0 \rho}} \frac{B_0}{2} \frac{Z_{\text{eff}}}{Z_{\text{eff}} + 1} \vec{\nabla} \tilde{p}(\tilde{R}) = -\vec{\nabla} \varphi.\end{aligned}\quad (7)$$

The energy density of this electric field varies as fourth power of B_0 . The condition that the energy density of electric field be negligible as compared with that of the magnetic field amounts to $V_A^2 \ll c^2$, which is met in many experimental situations, validating the neglect of electron pressure in Turner's theory.

The associated charge density is

$$\sigma = \varepsilon_0 \vec{\nabla} \cdot \vec{E} = \varepsilon_0 \frac{\omega_p}{c} \vec{\nabla} \cdot \vec{E} = \frac{1}{2} \frac{Z_{\text{eff}}}{Z_{\text{eff}} + 1} \varepsilon_0 B_0 \frac{e B_0}{m} \vec{\nabla}^2 \tilde{p}(\tilde{R}). \quad (8)$$

These results follow irrespective of the choice of the curl eigenfunctions $\vec{Z}_{\pm}(\tilde{\mu}_{\pm}, \tilde{R})$ or the specific geometry so long as the conditions of Turner's derivation, discussed in the last section, are satisfied.

III. CHOICE OF EIGENFUNCTION AND STRUCTURE OF TRS

The objective of seeking a model for the long-lived structures in Z-pinch provides some guidelines for seeking particular eigenfunctions of curl. Most experimentally observed long-lived structures have either a spheroidal form or a filamentary form. Accordingly, only axisymmetric eigenfunctions need to be considered. Also, Turner's general solution has two-scale lengths corresponding to the two eigenvalues. These imply a "structure within a structure." Experimental data about long-lived structures is insufficient to warrant any attention to a two-scale-length solution.

The simplest choice is to assume $\mu_- = 0$ for which, the eigenfunction is irrotational and therefore a harmonic function. In a simply connected domain, a harmonic field with vanishing normal at the boundary must be zero,²¹ thus one should assume $C_- = 0$. In order to generate numerical results, eigenfunctions of the curl operator in cylindrical geometry²² defined over a *notional* cylinder of radius "a" and height "b" are used. The eigenfunctions of $\vec{\nabla}$ corresponding to eigenvalue $\tilde{\mu}$ are of the form

$$\begin{aligned}\vec{Z}_{\tilde{\mu}} &= Z_{\tilde{\mu}r} \hat{r} + Z_{\tilde{\mu}\theta} \hat{\theta} + Z_{\tilde{\mu}z} \hat{z}, \\ Z_{\tilde{\mu}r} &= \left(\frac{P}{\tilde{\mu}} \right) \frac{j_p \pi}{\tilde{a} b} J_1(j_p \xi) \sin(P \zeta), \\ Z_{\tilde{\mu}\theta} &= \frac{j_p}{\tilde{a}} J_1(j_p \xi) \cos(P \zeta), \\ Z_{\tilde{\mu}z} &= \frac{1}{\tilde{\mu}} \left(\frac{j_p}{\tilde{a}} \right)^2 J_0(j_p \xi) \cos(P \zeta), \\ \xi &\equiv \frac{\tilde{r}}{\tilde{a}}; \quad \zeta \equiv \frac{\pi \tilde{z}}{\tilde{b}}; \quad \tilde{\mu}^2 = \left(\frac{j_p}{\tilde{a}} \right)^2 + \left(\frac{P \pi}{\tilde{b}} \right)^2,\end{aligned}\quad (9)$$

p: positive integers, P: odd integers.

Here, j_p is the pth zero of Bessel function of the first kind of order 1. Turner's derivation assumes the boundary condition that the normal components of $\vec{B}$ and $\vec{v}$ vanish at the boundary, which is already incorporated in Eq. (9). It is reasonable to expect that these components vanish *only* at the external boundary: there are no "boundaries within the boundary." This further suggests that $p=1$, $j_1=3.83171$, $P=1$. The axial current density should carry a total current I within its first zero, at $\xi_1=0.627612$ in the midplane, which leaves the model with the following free parameters: total current I, density n, radius a, and height b.

The above results then take on a simple form after choosing $C_+ = (1/2) \tilde{\mu}_+ \{ \tilde{\mu}_+ - \sqrt{(\tilde{\mu}_+)^2 + 4} \}$

$$\vec{B} = B_0 \vec{Z}, \quad (10)$$

$$\vec{J} = B_0 \frac{\omega_p}{\mu_0 c} \tilde{\mu} \vec{Z}, \quad (11)$$

$$\begin{aligned}\vec{v} &= -\frac{1}{2} \frac{c}{\omega_p} \frac{e B_0}{m} \vec{Z} \{ \sqrt{\tilde{\mu}^2 + 4} - \tilde{\mu} \} \\ &= -\frac{1}{2} \frac{B_0}{\sqrt{\mu_0 \rho}} \vec{Z} \{ \sqrt{\tilde{\mu}^2 + 4} - \tilde{\mu} \},\end{aligned}\quad (12)$$

$$\vec{E} = \frac{B_0}{\sqrt{\mu_0 \rho}} \frac{B_0}{2} \frac{Z_{\text{eff}}}{Z_{\text{eff}} + 1} \frac{1}{4} \{ \tilde{\mu} - \sqrt{\tilde{\mu}^2 + 4} \}^2 \vec{\nabla} |\vec{Z}|^2, \quad (13)$$

$$p(\tilde{R}) + \frac{|\vec{B}|^2}{2\mu_0} \frac{1}{4} \{ \tilde{\mu} - \sqrt{\tilde{\mu}^2 + 4} \}^2 = p_0.$$

These equations describe a force-free magnetic field and a simple Beltrami flow, symmetric about the $z=0$ plane and the z -axis. The new aspect introduced by Turner's theory is the relation between the magnetic field and the velocity field through Eq. (12) and pressure and magnetic field through Eq. (13). Earlier, Sudan has also obtained finite-pressure equilibria with mass flow velocity parallel to magnetic field and a Bernoulli condition similar to Eq. (4), treating total mass, canonical momenta and magnetic helicity as constant rather than hybrid helicity and magnetic helicity.¹⁰

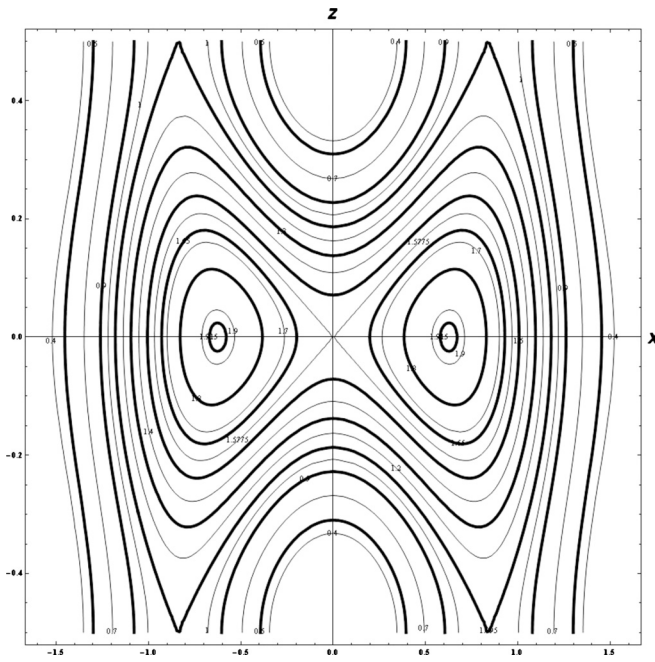
Several striking features are revealed by these expressions. The magnetic field amplitude is related to the axial current I flowing through the midplane of this structure as

$$B_0 = 0.8 \frac{\mu_0 I \omega_p}{2\pi c}. \quad (14)$$

The kinetic energy of ions deduced from Eq. (12) is

$$\begin{aligned}E_i &\equiv \frac{1}{2} m |\vec{v}|^2 \sim 6 \times 10^5 \cdot I^2 (\text{MA}) |\vec{Z}|^2 \\ &\times \frac{1}{4} (\sqrt{\tilde{\mu}^2 + 4} - \tilde{\mu})^2 \text{ (eV)}.\end{aligned}\quad (15)$$

The ratio of kinetic energy density to magnetic energy density is

FIG. 1. Contour map of $|\vec{Z}|^2$, modulus-squared of the curl eigenfunction.

$$\frac{\frac{1}{2}\rho|\vec{v}|^2}{|\vec{B}|^2/2\mu_0} = \frac{1}{4}\{\sqrt{\tilde{\mu}^2 + 4} - \tilde{\mu}\}^2. \quad (16)$$

The kinetic energy density and the magnetic energy density become almost equal when $\tilde{\mu} \ll 2$. For a given aspect ratio $\epsilon \equiv b/a$, this implies $a \gg (1/2)c\omega_p^{-1}\sqrt{(j_1^2 + \pi^2 z^2)}$ ($a \gg 2.477c\omega_p^{-1}$ for $\epsilon=1$). For this condition, the pressure balance equation *formally* becomes similar to a diamagnetic pressure balance equation, although it is in fact a Bernoulli equilibrium:

$$p(\vec{R}) + \frac{1}{2}\rho|\vec{v}(\vec{R})|^2 = p(\vec{R}) + \frac{|\vec{B}(\vec{R})|^2}{2\mu_0} = \frac{B_t^2}{2\mu_0} \Big|_{\text{boundary}}. \quad (17)$$

This uses the fact that since the *normal* components of magnetic field vanish at the boundary, p_0 must be related to the tangential magnetic field B_t at the *zero-pressure boundary*, which may not be the same as boundaries of the cylindrical box over which the eigenfunction is defined.

These features distinguish the TRS from the spheromak configuration. It is a force-free magnetic field with *finite pressure* supported by the magnetic pressure at the zero-pressure boundary *having equipartition between kinetic energy and magnetic energy*. The pressure is actually confined by a sheared velocity field through a Bernoulli equilibrium rather than through a magnetic force. Stability of plasma equilibria maintained by sheared velocity fields has been studied by Ohsaki and Yoshida.¹⁸ Similar conclusions are obtained in Ref. 17 where the Beltrami fields are *postulated* as steady state solutions to vorticity evolution equations rather than derived from a variational principle.

The modulus-squared of the eigenvector $|\vec{Z}|^2$ governs the spatial distribution of the pressure, electric potential, the Ohmic power dissipation density $\eta|\vec{J}|^2$, the kinetic energy

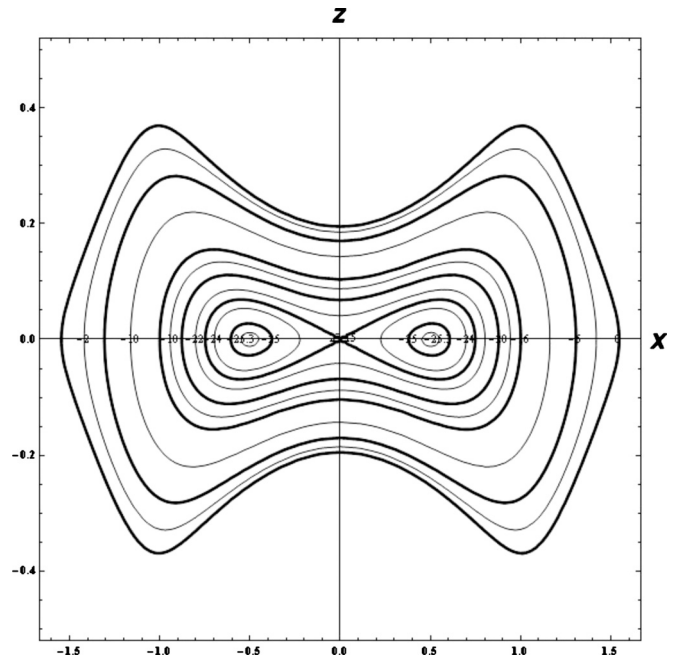


FIG. 2. Contour map of charge density.

density and the magnetic energy density, hence the entire shape and structure of the TRS. Figure 1 shows a contour-map of $|\vec{Z}|^2$ for $\tilde{a}=\tilde{b}=1$, over the interval $-1.6\tilde{a} \leq \tilde{x} \leq 1.6\tilde{a}$; $-0.5\tilde{b} \leq \tilde{z} \leq 0.5\tilde{b}$. Three different topologies are seen to coexist: enclosed domains of two kinds and one open domain. The innermost enclosed domain consists of nested toroids. This is surrounded by an orange-shaped simply connected enclosed domain. This is surrounded by a domain which has open boundaries within the displayed interval.

The charge density is seen to be proportional to $\tilde{v}^2|\vec{Z}|^2$. Figure 2 shows the contour map of $\tilde{v}^2|\vec{Z}|^2$.

The structure is negatively charged: the outermost contour is of zero charge density. The total charge on the structure scales with the aspect ratio as

$$Q = \frac{1}{2} \frac{Z_{\text{eff}}}{Z_{\text{eff}} + 1} \frac{1}{4} \{\tilde{\mu} - \sqrt{\tilde{\mu}^2 + 4}\}^2 \epsilon_0 B_0 \frac{eB_0}{m} \left(\frac{c}{\omega_p} \right) \pi a^2 F(\epsilon). \quad (18)$$

The function $F(\epsilon)$ is shown in Fig. 3. It is natural to inquire whether this is compatible with plasma quasi-neutrality. Quasineutrality would imply that this total charge represents a small fraction of the charge of all electrons within the cylinder which is $(en\pi a^2 b)$. This yields the inequality

$$\frac{1}{2} \frac{Z_{\text{eff}}}{Z_{\text{eff}} + 1} \frac{1}{4} \{\tilde{\mu} - \sqrt{\tilde{\mu}^2 + 4}\}^2 \frac{V_A^2}{c^2} \left(\frac{c}{\omega_p a} \right) \frac{|F(\epsilon)|}{\epsilon} \ll 1 \quad (19)$$

which is easily satisfied in most experimental situations.

The flow-line of the vector field $\vec{Z}$ passing through a given point (r_0, θ_0, z_0) described by the parametric equation

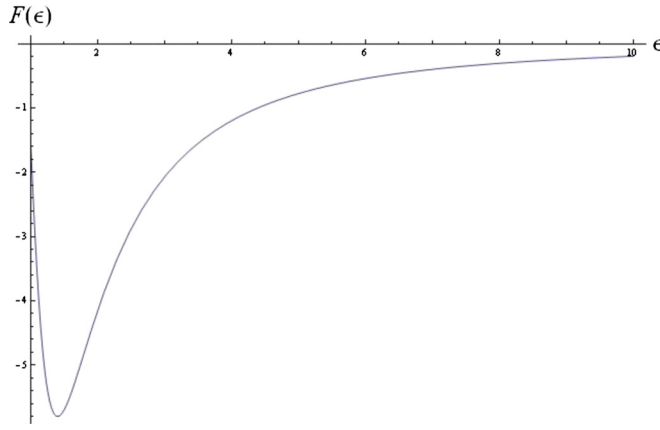


FIG. 3. (Color online) Scaling factor of total charge on the TRS as a function of aspect ratio ϵ .

$$\frac{dr}{Z_r} = \frac{rd\theta}{Z_\theta} = \frac{dz}{Z_z} = \frac{ds}{|\vec{Z}(r_0, \theta_0, z_0)|} \quad (20)$$

is shown in Fig. 4. It may be noted that this also represents the velocity field, which, in view of its extremely high particle kinetic energy seen in Eq. (15), can be identified with a trapped, circulating energetic ion beam.

The above discussion reveals the *mathematical object* referred as TRS to be simultaneously a fluid vortex, a force-free magnetic field, confined finite-pressure plasma, a charged structure and a trapped energetic ion beam. There is one difficulty associated with its adoption as a model for phenomena where these aspects are believed to play a significant role. This relates to the role played by the boundary of the notional cylinder over which the eigenfunctions are defined: it is difficult to visualize the physicality of this boundary inside a plasma focus or Z-pinch.

This difficulty may be circumvented by adapting the Gibbs ensemble approach to strong Hall MHD turbulence suggested by Servidio, Matthaeus, and Carbone.²³ In this approach, the plasma dynamical equations are expressed as a spectral expansion over a set of plane waves and the resulting Fourier–Galerkin modes are assumed to obey equilibrium statistics, with a probability density given by a generalized Gibbs canonical distribution function. Spectra of conserved quantities in the resulting theory are shown to have peaks at the long and short wavelength limits. Cylindrical curl eigenfunctions, which form a complete orthogonal basis set,²¹ could be employed in an analogous manner instead of plane waves. Sizes and shapes of these eigenfunctions would be determined not by physical boundaries but the spectral properties of the resulting theory. This approach could possibly explain the occurrence of long-lived filamentary structures, plasma points, and spheroidal structures under differing plasma conditions.

Diagnostic features expected from the TRS hypothesis are discussed in the next section.

IV. DIAGNOSTIC FEATURES

This section discusses the scenario expected if the hypothetical TRS structure were to be physically realized in an

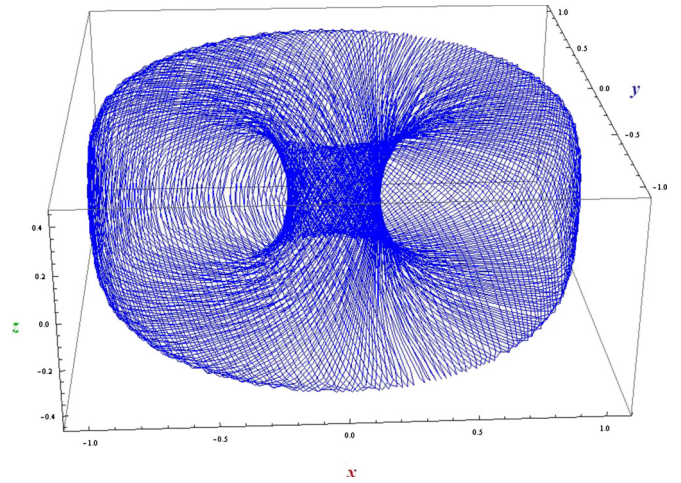


FIG. 4. (Color online) Three-dimensional flow-line of the axisymmetric curl eigenfunction.

experimental situation, in which incompressible and irrotational flows could coexist. Turner’s assumption of constant density could then be *approximately* valid for adiabatic spatial and temporal variations of density. The incompressible component of flow would then be expected to evolve toward a TRS while the irrotational flow would evolve toward a Boltzmann distribution, $n \cong n_0 \exp(-e\phi/kT)$, with ϕ following spatial distribution of $|\vec{Z}|^2$ (Fig. 1). It is interesting to observe features resembling “two eyes” in Fig. 5 of Ref. 3 in the interferogram corresponding to 180 ns at $z \sim 2.7$ cm and also in the “empty” region at $z \sim 0.45$ cm. Figure 13(a) of Ref. 2, dealing with simulation of fast ion generation during $m=0$ instability of a Z-pinch also has a feature similar to two eyes. The similarity of these features with Fig. 1 is probably fortuitous but noteworthy.

In this scenario, the plasma ion component associated with irrotational flow would represent a comparatively stationary target being hit by a toroidal stream of plasma ions from the incompressible flow with high kinetic energy. The expected space-resolved neutron energy distribution in the y - z plane at a location (y_0, z_0) can be calculated as follows. The energy of the neutron seen by the detector would be

$$E_{n,\hat{x}} = \frac{1}{2} m_n (v_{n,CM} + \frac{1}{2} \vec{v} \cdot \hat{x})^2. \quad (21)$$

Assuming that the detector has an energy resolution of ΔE and a spatial resolution $\Delta y, \Delta z$ in y and z directions respectively, the reaction rate $\sigma(|\vec{v}|)|\vec{v}|$ can be calculated on a lattice of points filling the region $-a \leq x \leq a$; $y_0 - (1/2)\Delta y \leq y \leq y_0 + (1/2)\Delta y$; $z_0 - (1/2)\Delta z \leq z \leq z_0 + (1/2)\Delta z$. The calculated reaction rate values can be added for those lattice points for which the energy according to Eq. (21) lies in a band $(E, E + \Delta E)$ to yield a representative spectrum. Figure 5 illustrates the result for $\Delta E = 0.05$ MeV, $\Delta y = \Delta z = 0.05$ cm:

There is an asymmetry in the fusion product (neutron or proton) spectra at mirror-symmetric y -coordinates, which has been experimentally observed^{5,6} and ascribed to a rotational motion of the reacting deuteron population. Space-integrated spectrum obtained using a similar exercise would show an abnormally high deuteron temperature, such as reported in

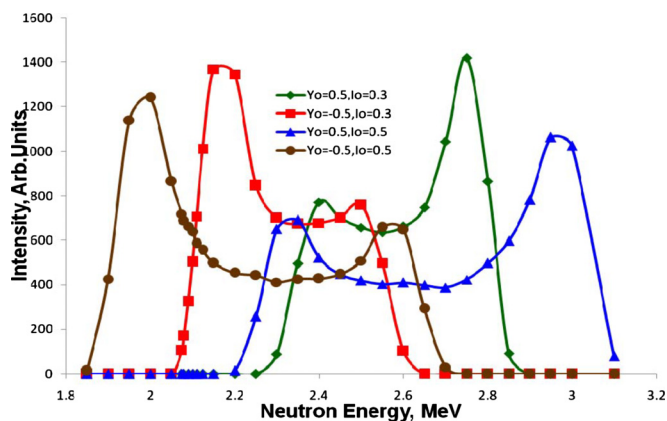


FIG. 5. (Color online) Space-resolved fusion-product energy spectrum at mirror-symmetric lateral coordinates for 0.3 and 0.5 MA trapped current arising from the TRS hypothesis.

Ref. 1. The comparison of fusion-proton diagnostics data on POSEIDON plasma focus with the Gyating Particle Model⁶ indicates a deuteron quasi-temperature in the range of 200–400 keV for a pinch current of 0.8 MA. Estimate of the scale of deuteron kinetic energy for this case based on Eq. (15) yields the value 385 keV. Detectors placed along the +z and –z directions would be expected to observe a shift of the neutron spectrum to higher and lower energy, resembling emission from an axial beam, which is observed quite commonly in plasma foci and Z-pinch. The TRS model indicates that a small population of neutrons observed *along the* –z direction should be shifted to *higher* energy corresponding to interactions near the outer boundary of the toroidal flow. Anisotropy of fusion reaction in terms of yield and reaction product spectrum depends on the aspect ratio and may account for much higher anisotropy in some Z-pinch² as being due to high aspect ratio TRS.

V. SUMMARY AND CONCLUSION

Properties of Turner's relaxed state solution for a bounded Hall MHD plasma have been studied in this paper from the point of view of ascertaining feasibility of constructing models of long-lived plasma structures observed in plasma focus and Z-pinch and conjectured to be associated with an ion-trapping mechanism responsible for the observed neutron emission. Assuming that only one scale-length exists in a practically interesting case, the TRS turns out to be simultaneously a fluid vortex, a force-free magnetic field, confined finite-pressure plasma, a charged structure and a trapped energetic ion beam. The difficulty of identifying a physical boundary defining the TRS suggests that TRS may be thought of as statistically preferred state of a Gibbs ensemble formulation of a general turbulence problem with

cylindrical curl eigenfunctions forming a basis set for the spectral expansion.

Diagnostic signatures expected from the TRS model are actually observed in some experiments. This suggests that further work on investigating the TRS model may be fruitful. More extensive experimental data on space-resolved neutron and fusion-proton spectra from laterally mirror-symmetric points on a dense plasma focus device and neutron spectral width determined as a function of observation angle would enable a detailed examination of the predictions of this model. Equilibrium statistical models of turbulence with cylindrical curl eigenfunctions as basis functions for spectral expansion may indicate the tendency for condensation of energy into a few eigenmodes which could correspond to Turner's two-eigenvalue solution⁹ or the double Beltrami states of Yoshida *et al.*¹⁷ or the triple-Beltrami solutions of Dasgupta *et al.*¹⁹ under various conditions.

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